

1. Install Docker Desktop or Docker Engine on your local machine and verify the installation by running the command `docker --version` in your terminal.

ANS:

- I installed Docker Docker Engine on my local machine. After installation, I opened the terminal and verified the Docker installation using:

- `docker --version`

❖ Expected Output:

- Docker version 28.x.x, build xxxxxxxx
- This confirms that Docker has been installed successfully and is available from the terminal.

2. Pull the official Nginx Docker image from Docker Hub using the command `docker pull nginx` and list all downloaded images with `docker images`.

ANS:

- I pulled the official Nginx image from Docker Hub using the following command

- `docker pull nginx`

- After downloading the image, I listed all the available Docker images using:

- `docker images`

- The output displayed the downloaded Nginx image.

❖ Expected Output:

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nginx	latest	xxxxxxx	...	...

- This confirms that the Nginx Docker image was successfully downloaded.

3. Start a new Nginx container using the command `docker run -d -p 8080:80 --name mynginx nginx`, then open your browser and check if you can access the default Nginx welcome page at <http://localhost:8080>.

ANS:

- I started a new Nginx container using:
 - `docker run -d -p 8080:80 --name mynginx nginx`
- ❖ Here:
 - `-d` runs the container in detached mode.
 - `-p 8080:80` maps host port 8080 to container port 80.
 - `--name mynginx` gives the container the name mynginx.
 - `nginx` specifies the Docker image to use.
- I then opened a web browser and entered:
 - `http://localhost:8080`
- To verify that the container was running, I used:
 - `docker ps`

❖ Expected Output:

CONTAINER ID	IMAGE	COMMAND	PORTS	NAMES
xxxxxxx	nginx	"/docker-entrypoint..."	0.0.0.0:8080->80/tcp	mynginx

4. Create a Docker volume named `zomato_data` using `docker volume create` and attach it to a new container running the `httpd` (Apache) image so that any data written to `/usr/local/apache2/htdocs` inside the container is stored persistently.
Hint: Use the `-v` flag with `docker run` to mount the volume.

ANS:

- I created a Docker volume named `zomato_data` using:
- I verified the volume using:
 - `docker volume ls`
- Then I created an Apache HTTPD container and mounted the volume to the Apache web root:

```
docker run -d \
--name zomato-apache \
-p 8081:80 \
-v zomato_data:/usr/local/apache2/htdocs \
Httpd
```

5. Compare containers and virtual machines by listing 3 benefits of Docker containers over traditional VMs, using examples from apps like Zomato, Instagram, or Flipkart.

ANS:

Benefit	Docker Containers	Example
1. Lightweight	Containers share the host OS kernel, so they require fewer resources than traditional VMs.	Zomato can run many small services without requiring a separate full OS for each service.
2. Fast Startup	Containers can start within seconds because they do not need to boot a complete operating system.	Instagram can quickly start application containers when additional capacity is required.
3. Easy Deployment and Portability	Docker containers package the application and its dependencies together, making deployment consistent across environments.	Flipkart developers can test an application in Docker locally and deploy the same container to a cloud server.